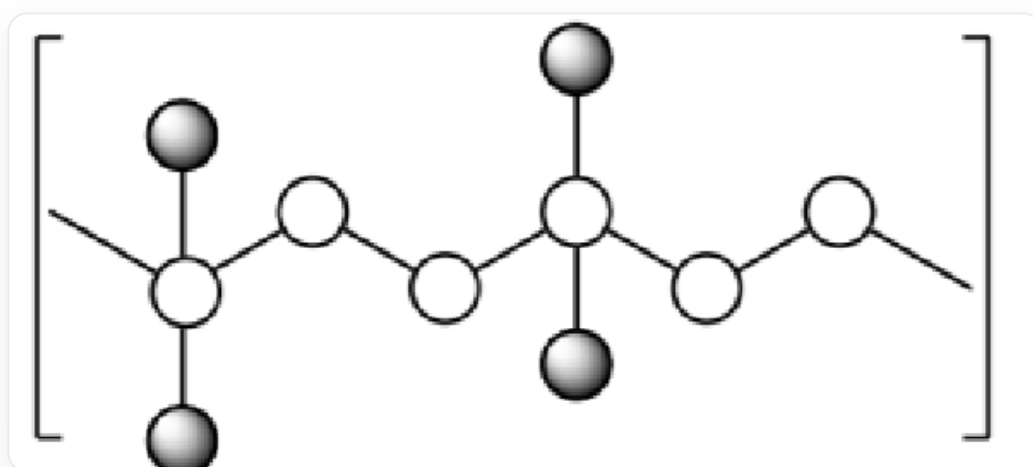


Question

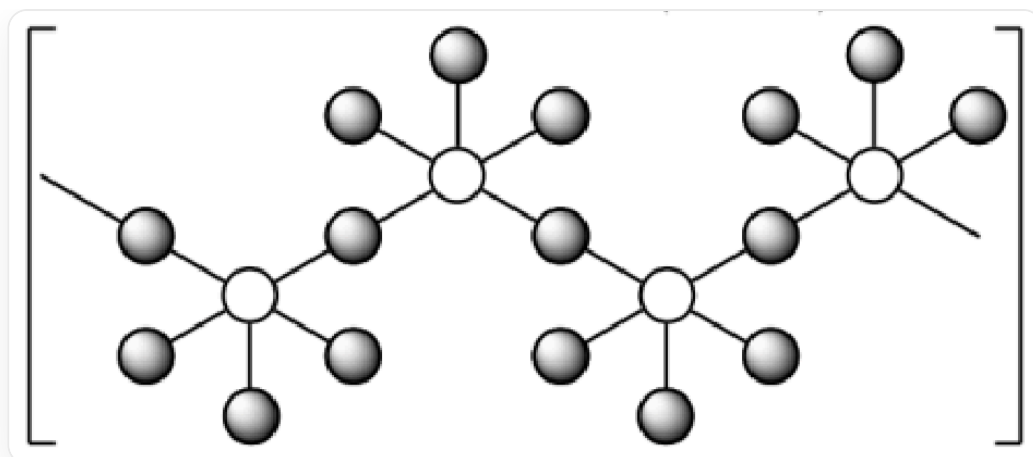
The non-radioactive element **A** can form a series of binary polymers with chain-like structures when combined with various halogens. The schematic structures are as follows (hollow/white spheres represent element **A**, solid/black spheres represent halogens), where the mass fractions of element **A** in **X1** to **X4** are 84.37%, 62.67%, 76.16%, and 50.14%, respectively:

- Image for **X1**



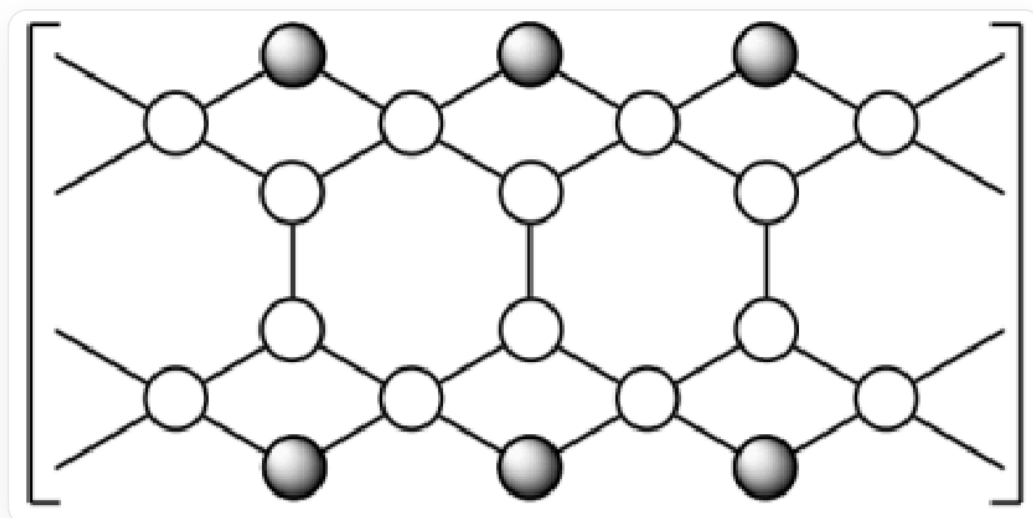
In the image, hollow/white spheres are linked to form a one-dimensional long chain similar to the carbon chain in polyethylene. If the one-dimensional chains are numbered 1, 2, 3..., then every hollow/white sphere numbered $3N$ is connected to two solid/black spheres as terminal groups.

- Image for **X2**



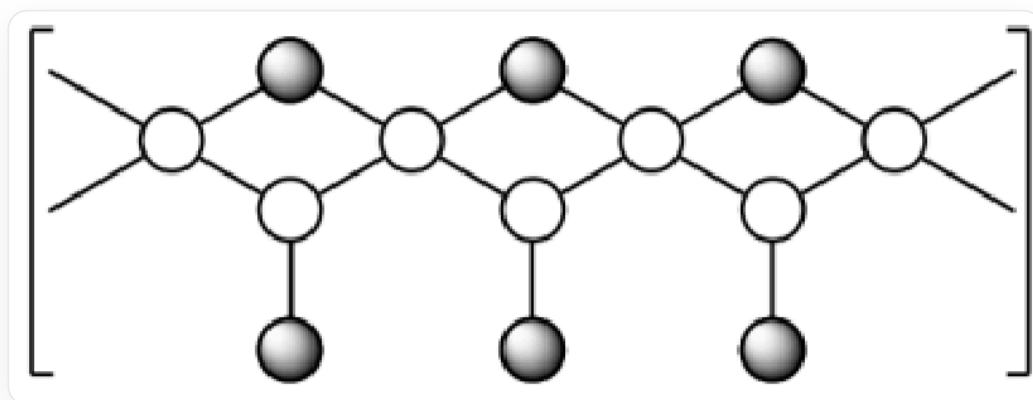
In the image, hollow/white spheres and solid/black spheres are alternately linked to form a one-dimensional long chain. Each hollow/white sphere is connected to five solid/black spheres, three of which are terminal groups and two are bridging.

- Image for X3



In the image, hollow/white spheres form long chains of fused hexagon rings sharing edges, resembling the fusion of benzene rings in polyacenes. For every two hollow/white spheres on the same side of a hexagon that are not involved in edge-sharing, an additional solid/black sphere bridges them, forming an extra four-membered ring structure. The hollow/white spheres not involved in edge-sharing hexagons are connected to two hollow/white spheres that are part of edge-sharing hexagons and to two bridging solid/black spheres forming the four-membered ring.

- Image for X4



In the image, hollow/white spheres form the same one-dimensional long chain as in **X1**. If the chain is numbered 1, 2, 3..., adjacent odd-numbered hollow/white spheres are bridged by an additional solid/black sphere to form a four-membered ring. Every odd-numbered hollow/white sphere has two bridging solid/black spheres, while even-numbered hollow/white spheres are connected to one terminal solid/black sphere.

Based on the information about the inorganic chain-like polymers in the images, deduce the stoichiometric ratio of **A** atoms to halogen atoms in **X1-X4**, and calculate to determine which element **A** is and the chemical formulas of **X1-X4**. Then, identify the number of correct propositions from the following:

1. Element **A** belongs to Group 15.
2. Element **A** is located in the fifth period.
3. When the halogen atoms in **X1-4** are ordered by their periodic table positions from low to high, the sequence is **X1**, **X2**, **X4**, **X3**.
4. The ratio of **A** atoms to halogen atoms in **X1-4**, ordered from smallest to largest, is **X2**, **X4**, **X3**, **X1**.
5. According to hybrid orbital theory, **A** atoms in **X1** exhibit two hybridization modes, and their ratio is 1:1.
6. According to hybrid orbital theory, **A** atoms in **X2** are sp^3d^2 hybridized.
7. **X3** contains two types of **A** atoms with different chemical situations, and their numbers are equal.

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. 7

Answer

Correct Answer: **D**

Detailed Explanation

First, based on the problem description and structural diagrams, determine the simplest integer ratio of **A** atoms to halogen atoms (**X**) for the four polymers (**X1**, **X2**, **X3**, **X4**).

* **X1**: The structure consists of a long chain of **A** atoms, where every third **A** atom (after every two **A** atoms) is bonded to two terminal halogen atoms. That is, $\dots - \text{A} - \text{A} - \text{A}(\text{X}_2) - \text{A} - \text{A} - \text{A}(\text{X}_2) - \dots$. Thus, the repeating unit is A_3X_2 .

* **A:X ratio = 3:2**

CHECKPOINT

1 PTS

The simplest formula for **X1** is A_3X_2

X2: Each **A** atom is bonded to five halogen atoms, three of which are terminal and two are bridging. The bridging halogen atoms are shared between two **A** atoms. Therefore, the effective number of halogen atoms per **A** atom is $3 + 2(1/2) = 4$. The repeating unit is AX_4 .

* **A:X ratio = 1:4**

CHECKPOINT

1 PTS

The simplest formula for **X2** is AX_4

* **X3**: The structure is a ladder-like chain composed of fused six-membered rings. From the diagram, the smallest repeating unit contains four **A** atoms (two on the inner side of the "ladder" and two on the outer side) and two bridging halogen atoms. Thus, the repeating unit is A_4X_2 , and the simplest formula is A_2X .

* **A:X ratio = 2:1**

CHECKPOINT

1 PTS

The simplest formula for **X3** is A_2X

X4: The structure is a long chain of **A** atoms, where **A** atoms at even positions are bonded to one terminal halogen atom, and **A** atoms at odd positions are bridged by one halogen atom. Considering an $A(\text{odd})$ - $A(\text{even})$ repeating unit: $A(\text{even})$ has one terminal halogen; $A(\text{odd})$ is connected to two bridging halogen atoms, each contributing $1/2$, totaling $2(1/2) = 1$ halogen atom. Thus, two **A** atoms correspond to two halogen atoms. The repeating unit is A_2X_2 , and the simplest formula is AX .

* **A:X ratio = 1:1**

CHECKPOINT

1 PTS

The simplest formula for **X4** is AX

2. Inferring Element **A** and the Chemical Formulas of the Species

Let the atomic weight of element **A** be M_A , and the atomic weight of the halogen be M_X . Using the mass fraction formula $w_A = \frac{n_A \cdot M_A}{n_A \cdot M_A + n_X \cdot M_X}$, we can derive $M_A = \frac{w_A \cdot n_X \cdot M_X}{n_A \cdot (1 - w_A)}$.

We match the four stoichiometric ratios and mass fractions, testing possible halogens (F=19.0, Cl=35.5, Br=79.9, I=126.9) to solve for a consistent M_A .

* **For the formula AX_4 (A:X = 1:4), mass fraction $w_A = 62.67\%$:**

$$M_A = \frac{0.6267 \cdot 4 \cdot M_X}{1 \cdot (1 - 0.6267)} \approx 6.716 \cdot M_X.$$

If **X** is **F** (fluorine), $M_A \approx 6.716 \cdot 19.0 = 127.6$.

* **For the formula A_3X_2 (A:X = 3:2), mass fraction $w_A = 84.37\%$:**

$$M_A = \frac{0.8437 \cdot 2 \cdot M_X}{3 \cdot (1 - 0.8437)} \approx 3.601 \cdot M_X.$$

If **X** is **Cl** (chlorine), $M_A \approx 3.601 \cdot 35.5 = 127.8$.

* **For the formula A_2X (A:X = 2:1), mass fraction $w_A = 76.16\%$:**

$$M_A = \frac{0.7616 \cdot 1 \cdot M_X}{2 \cdot (1 - 0.7616)} \approx 1.597 \cdot M_X.$$

If **X** is **Br** (bromine), $M_A \approx 1.597 \cdot 79.9 = 127.6$.

* **For the formula AX (A:X = 1:1), mass fraction $w_A = 50.14\%$:**

$$M_A = \frac{0.5014 \cdot 1 \cdot M_X}{1 \cdot (1 - 0.5014)} \approx 1.0056 \cdot M_X.$$

If **X** is **I** (iodine), $M_A \approx 1.0056 \cdot 126.9 = 127.6$.

All calculations point to $M_A \approx 127.6$, identifying the element as **tellurium (Te)**. It is a non-radioactive element, consistent with the problem statement.

CHECKPOINT

2 PTS

Element **A** is **Te**

Thus, the chemical formulas of the four species are:

* **X1:** Te_3Cl_2

* **X2:** TeF_4

* **X3:** Te_2Br

* **X4:** TeI

CHECKPOINT

1 PTS

X1-X4 are Te_3Cl_2 , TeF_4 , Te_2Br , TeI respectively

3. Evaluating the Truth of Propositions

Now, we analyze the eight given propositions one by one:

1. Element A belongs to Group V (15)

Tellurium (Te) is located in Group VI (16, the chalcogen group) of the periodic table, not Group V (15, the pnictogen group). Thus, this proposition is **false**.

2. Element A is in the fifth period

The valence electron shell of tellurium (Te) is the fifth shell ($n=5$), placing it in the fifth period. Thus, this proposition is **true**.

CHECKPOINT

1 PTS

Tellurium (Te) is in Group VI, Period 5. Proposition 1 is false, 2 is true

3. When ordering the halogen atoms in X1-4 by increasing period number, the sequence is X1, X2, X4, X3

* X1: Cl (chlorine, Period 3)

* X2: F (fluorine, Period 2)

* X3: Br (bromine, Period 4)

* X4: I (iodine, Period 5)

The correct order from low to high period is: F (X2) < Cl (X1) < Br (X3) < I (X4). The given sequence is X1, X2, X4, X3. Thus, this proposition is **false**.

CHECKPOINT

1 PTS

F (X2) < Cl (X1) < Br (X3) < I (X4). Proposition 3 is false

4. When ordering the A:halogen atom ratio in X1-4 from smallest to largest, the sequence is X2, X4, X3, X1

* X1 (Te₃Cl₂): A:X = 3:2 = 1.5

* X2 (TeF₄): A:X = 1:4 = 0.25

* X3 (Te₂Br): A:X = 2:1 = 2.0

* X4 (TeI): A:X = 1:1 = 1.0

The correct order from smallest to largest is: 0.25 (X2) < 1.0 (X4) < 1.5 (X1) < 2.0 (X3). The given sequence is X2, X4, X3, X1. Thus, this proposition is **false**.

CHECKPOINT

1 PTS

Halogen atom ratios: 0.25 (X2) < 1.0 (X4) < 1.5 (X1) < 2.0 (X3). Proposition 4 is false

5. According to hybridization theory, X1 has two types of hybridized A atoms, with a 1:1 ratio

In the X1 (Te₃Cl₂) chain, there are two distinct tellurium environments:

* Te bonded to two other Te atoms: 2 bonding pairs and 2 lone pairs, sp³ hybridization.

* Te bonded to two Te atoms and two Cl atoms: Te(IV) with 4 bonding pairs and 1 lone pair (total 5 valence electron pairs), sp³d hybridization.

The ratio of these Te atoms is 2:1, not 1:1. Thus, this proposition is **false**.

CHECKPOINT

1 PTS

X1 has two distinct Te environments with a 2:1 ratio. Proposition 5 is false

6. According to hybridization theory, X2 has A atoms with sp^3d^2 hybridization

In **X2** (TeF_4), each Te atom bonds with five F atoms, and Te is in the +4 oxidation state with 1 lone pair. The total valence electron pairs are 5 (bonding) + 1 (lone) = 6. Six electron pairs correspond to sp^3d^2 hybridization. Thus, this proposition is **true**.

CHECKPOINT

1 PTS

X2 Te atoms are sp^3d^2 hybridized. Proposition 6 is true

7. X3 contains two chemically distinct A atoms in equal numbers

In **X3**'s ladder structure, there are two distinct Te environments:

- * Inner Te: bonded to 3 Te atoms.
- * Outer Te: bonded to 2 Te atoms and 1 Br atom.

Their counts are equal in the repeating unit (2 each). Thus, this proposition is **true**.

CHECKPOINT

1 PTS

X3 has two chemically distinct Te atoms in equal numbers. Proposition 7 is true

In summary, the correct propositions are 2, 6, and 7, totaling **3**.

The correct answer is D.